Course Name:

Course Outcome

CO1: Understand the classification of operating system environment.
CO2: Understand the basic of process management.
CO3: Apply the concept of CPU process scheduling for the given scenarios.
CO4: Illustrate the process synchronization and concurrency process in operating system.
CO5: Analyze the occurrence of deadlock in operating system.
CO6: Describe and analyze the memory management and its allocation policies.
CO7: Understand the concepts of disk scheduling.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Even Semester 2023-24
B.Tech (CSE) II Year, IV Semester
BCSC0004: Operating System

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

1. Attempt all questions
2. do not switch between section A and Section B; first, solve one section, then move to second
3. Use only a blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Imagine you're working with a group of non-technical colleagues who are curious about computers. While discussing operating systems, they ask, "Why do we even need an operating system in our computers?" How would you explain the main purpose and significance of an operating system in a relatable scenario or example?	3	1	R	F
2	Consider the following code for process P0 and P1: **Process P_0** ... **While (Flag[1]==true) ;** **Flag[0] =True** **<Critical Section>** **Flag[0]=False;** .... **Process P_1** ... **While (Flag[0]==true) ;** **Flag[1]=True** **<Critical Section>** **Flag[1]=False;** .... Find out whether this solution is satisfying all the requirement of critical section.	3	2	U	C
3	Consider a scenario where a user creates a new task. Describe the sequence of state transitions that would occur in the process state diagram, starting from the initial state and ending with the task being marked as	3	.2	R	F

	completed. Additionally, identify any decision points or conditions that might affect the transitions between states. Discuss your answer by considering that there is limited main memory in the system.				
4	How process is represented in the memory? Explain with an example. Also, discuss the importance of Process control block.	2	2	U	F
5	Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?	3	3	A	P

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	An operating system uses Round Robin CPU Scheduling with a time quantum of 2 ms. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds): Process Arrival Time Burst Time P1 0 7 P2 2 4 P3 3 6 P4 8 5 Find the average waiting time and turnaround time of the processes.	5	3	A	C
7	Design a solution for the reader-writer synchronization problem by considering library system. Outline the algorithm or approach you would use to implement this problem, considering the following conditions:	5	4	A	P

	1. Readers should be allowed to access the collection concurrently. 2. Writers must have exclusive access to modify the collection, meaning no other readers or writers can access it during a write operation. 3. Ensure fairness in accessing the collection, preventing starvation of either readers or writers.				
8	Discuss the usage of semaphore with operations? Also, explain the term busy waiting and how it is overcome?	5	4	U	F